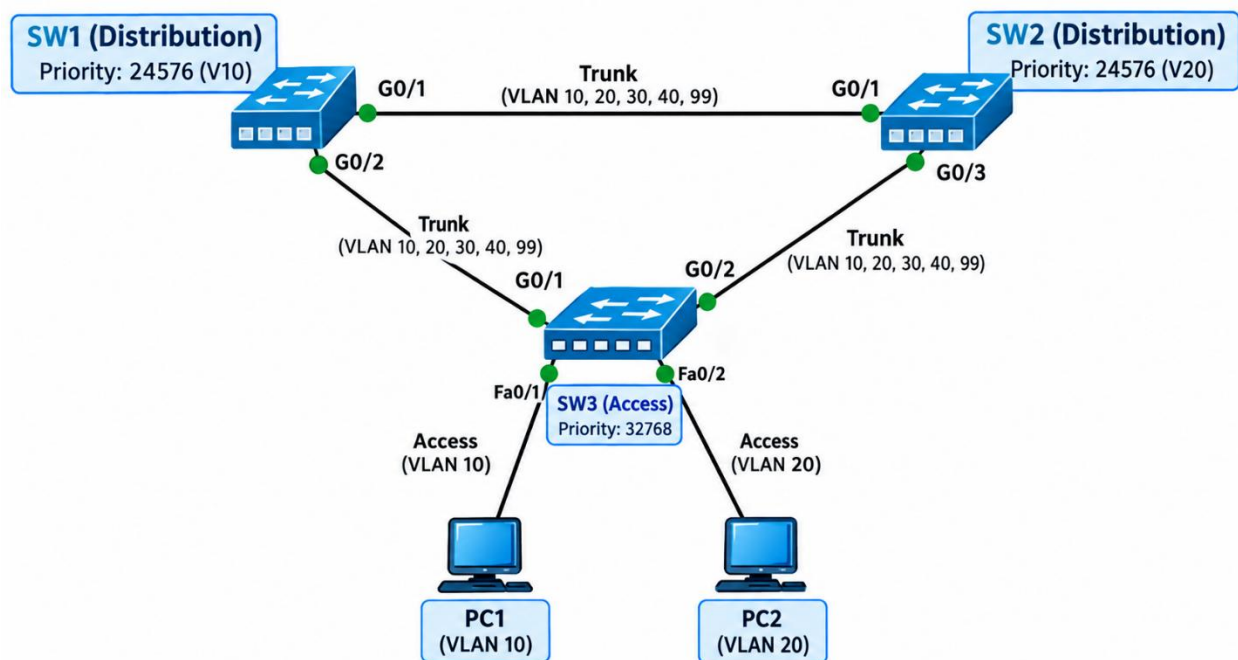


Rapid PVST+ Lab Core real network design : Per-VLAN Load Balancing & STP Optimization



1. Objectives

- Transition all switches from legacy STP/PVST to **Rapid PVST+** (spanning-tree mode rapid-pvst).
- Configure dual-root distribution:
 - **SW1**: Root Primary for VLANs 10, 30; Root Secondary for VLANs 20, 40, 99.
 - **SW2**: Root Primary for VLANs 20, 40; Root Secondary for VLANs 10, 30, 99.
- Manipulate Spanning Tree path cost to influence Root Port (RP) selection deterministically.

- Implement Layer-2 edge defense using **PortFast** and **BPDU Guard** on access ports.
- Deploy **Root Guard** on distribution-to-access links to prevent unauthorized root takeovers.
- Verify port roles (Root, Designated, Alternate) and states (Discarding, Forwarding) per VLAN.

VLAN ID	VLAN Name	Purpose	Subnet / Gateway
10	Engineering	Odd VLAN Data	192.168.10.0/24
20	Operations	Even VLAN Data	192.168.20.0/24
30	Development	Odd VLAN Data	192.168.30.0/24
40	Executive	Even VLAN Data	192.168.40.0/24
99	Management	Switch SVI	192.168.99.0/24

SVI & End-Host Addressing

Device	Interface	IP Address	Subnet Mask	Default Gateway
SW1	VLAN 99	192.168.99.11	255.255.255.0	N/A
SW2	VLAN 99	192.168.99.12	255.255.255.0	N/A
SW3	VLAN 99	192.168.99.13	255.255.255.0	192.168.99.1
PC1	NIC	192.168.10.10	255.255.255.0	192.168.10.1
PC2	NIC	192.168.20.10	255.255.255.0	192.168.20.1

2. Basic configuration steps

Apply on SW1, SW2, and SW3

enable

configure terminal

no ip domain-lookup

vlan 10

name ENGINEERING

vlan 20

name OPERATIONS

vlan 30

name DEVELOPMENT

vlan 40

name EXECUTIVE

vlan 99

```
name MANAGEMENT
```

```
exit
```

```
spanning-tree mode rapid-pvst
```

Configure management SVIs:

SW1

```
interface vlan 99
```

```
ip address 192.168.99.11 255.255.255.0
```

```
no shutdown
```

```
exit
```

SW2

```
interface vlan 99
```

```
ip address 192.168.99.12 255.255.255.0
```

```
no shutdown
```

```
exit
```

SW3

```
interface vlan 99
```

```
ip address 192.168.99.13 255.255.255.0
```

```
no shutdown
```

```
exit
```

Trunk Links

SW1

```
interface range GigabitEthernet 0/1 - 2
```

```
description TRUNK_INTERSWITCH
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99
no shutdown
exit
```

SW2

```
interface range GigabitEthernet 0/1 , GigabitEthernet 0/3
description TRUNK_INTERSWITCH
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99
no shutdown
exit
```

SW3

```
interface range GigabitEthernet 0/1 - 2
description UPLINK_TO_DISTRIBUTION
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99
no shutdown
exit
```

Per-VLAN Load Balancing

SW1 Configuration

```
configure terminal
spanning-tree vlan 10,30 priority 24576
```

```
spanning-tree vlan 20,40,99 priority 28672
exit
```

SW2 Configuration

```
configure terminal
spanning-tree vlan 20,40 priority 24576
spanning-tree vlan 10,30,99 priority 28672
exit
```

Access Port (SW3)

SW3 Access Ports

```
interface FastEthernet 0/1
description PC1_ENGINEERING
switchport mode access
switchport access vlan 10
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown
exit
```

```
interface FastEthernet 0/2
description PC2_OPERATIONS
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
```

no shutdown

exit

Global Configuration Loop Guard

(Root and Alternate roles)

Apply on SW1, SW2, and SW3

configure terminal

spanning-tree loopguard default

exit

Interface-Level Configuration(optional)

SW3 Configuration

configure terminal

interface range GigabitEthernet 0/1 - 2

description UPLINK_TO_DISTRIBUTION

spanning-tree guard loop

exit

STP Root Guard Deployment

SW1

interface GigabitEthernet 0/2

spanning-tree guard root

exit

SW2

interface GigabitEthernet 0/3

spanning-tree guard root

exit

STP Cost Tuning (Traffic Engineering Exercise)

By default, GigabitEthernet has an RSTP path cost of 20,000 (or 4 on short path cost standard).
Manually adjust the cost on SW3 for VLAN 10 to force uplink path asymmetry if required:

SW3: Increase cost of G0/1 specifically for VLAN 10

interface GigabitEthernet 0/1

spanning-tree vlan 10 cost 25000

exit

Verification

show spanning-tree summary

show spanning-tree root

show spanning-tree vlan 10

show spanning-tree vlan 20

show spanning-tree interface FastEthernet 0/1 detail

show spanning-tree inconsistentports

show interfaces status err-disabled

for recover from particular port from error disable state need to type shutdown no shutdown

interface FastEthernet 0/1

shutdown

no shutdown

Trigger Root Guard Inconsistency: On SW3, lower its priority for VLAN 10 to 4096 (spanning-tree vlan 10 priority 4096). Observe G0/2 on SW1 entering the ROOT_Inc (Root Inconsistent) state:

show spanning-tree inconsistentports

Restore SW3 priority back to default (32768) and verify the link self-heals back to forwarding without link flaps.

Full configuration for this lab

SW1

configure terminal

Enable global IPv4 routing

ip routing

Create SVI for VLAN 10 (Engineering Gateway)

interface vlan 10

description GATEWAY_ENGINEERING

ip address 192.168.10.1 255.255.255.0

no shutdown

Create SVI for VLAN 20 (Operations Gateway)

interface vlan 20

description GATEWAY_OPERATIONS

ip address 192.168.20.1 255.255.255.0

no shutdown

Create SVI for VLAN 30 (Development Gateway)

```
interface vlan 30  
description GATEWAY_DEVELOPMENT  
ip address 192.168.30.1 255.255.255.0  
no shutdown
```

Create SVI for VLAN 40 (Executive Gateway)

```
interface vlan 40  
description GATEWAY_EXECUTIVE  
ip address 192.168.40.1 255.255.255.0  
no shutdown  
exit  
enable  
configure terminal
```

1. Exclude the Gateway IPs

```
ip dhcp excluded-address 192.168.10.1  
ip dhcp excluded-address 192.168.20.1  
ip dhcp excluded-address 192.168.30.1  
ip dhcp excluded-address 192.168.40.1
```

2. Create DHCP Pool for VLAN 10 (Engineering)

```
ip dhcp pool VLAN10_POOL  
network 192.168.10.0 255.255.255.0  
default-router 192.168.10.1  
dns-server 8.8.8.8
```

3. Create DHCP Pool for VLAN 20 (Operations)

```
ip dhcp pool VLAN20_POOL  
network 192.168.20.0 255.255.255.0  
default-router 192.168.20.1  
dns-server 8.8.8.8
```

4. Create DHCP Pool for VLAN 30 (Development)

```
ip dhcp pool VLAN30_POOL  
network 192.168.30.0 255.255.255.0  
default-router 192.168.30.1  
dns-server 8.8.8.8
```

5. Create DHCP Pool for VLAN 40 (Executive)

```
ip dhcp pool VLAN40_POOL  
network 192.168.40.0 255.255.255.0  
default-router 192.168.40.1  
dns-server 8.8.8.8  
exit
```

Note: SW2 and SW3 No Configuration required

Verification command

```
show ip route
```

```
show ip dhcp pool
```

```
show ip dhcp binding
```